STM32-Based Quadcopter Flight Controller

PCB Version 1 — Technical Design Documentation

Hardware Design • Schematic • Power Architecture • PCB Layout • Routing • Manufacturing

Document information	Value
Design revision	Rev 0.1 – WIP (schematic title block)
PCB file	flight_controller_version1(4).kicad_pcb
Schematic file	flight_controller_version1.kicad_sch
PCB dimensions	64.10 × 50.50 mm
PCB thickness	1.5982 mm
Layer count	4 copper layers
Primary MCU	STM32F411CEUx
Primary IMU	ICM-20948
CAD tool	KiCad 10.0 file format
Documentation status	Engineering draft based on current design files

Scope: This document is a technical documentation draft generated from the currently uploaded KiCad schematic and PCB. It intentionally distinguishes verified design data from engineering interpretation and unresolved information.

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1. Executive Summary

The project is a custom embedded flight-controller PCB centered on an STM32F411 microcontroller and an ICM-20948 inertial measurement unit. The current KiCad PCB is a four-layer board with dedicated internal copper layers, multiple regulated voltage domains, USB, SWD, RC/serial interfaces, and four PWM-labelled control signals.

The design contains a dedicated switching regulator based on the TPS54336ADDA, an AMS1117-3.3 linear regulator, a TLV75518PDBV 1.8 V regulator, an AO3401A MOSFET, a 120 Ω ferrite bead, and a set of SN74AUP1T34DCK single-bit logic buffers. The design also includes a 16 MHz crystal, USB Micro-B, SWD, an XT60 battery connector, and multiple three-/four-pin peripheral connectors.

2. System Architecture

The PCB can be divided into the following functional blocks:

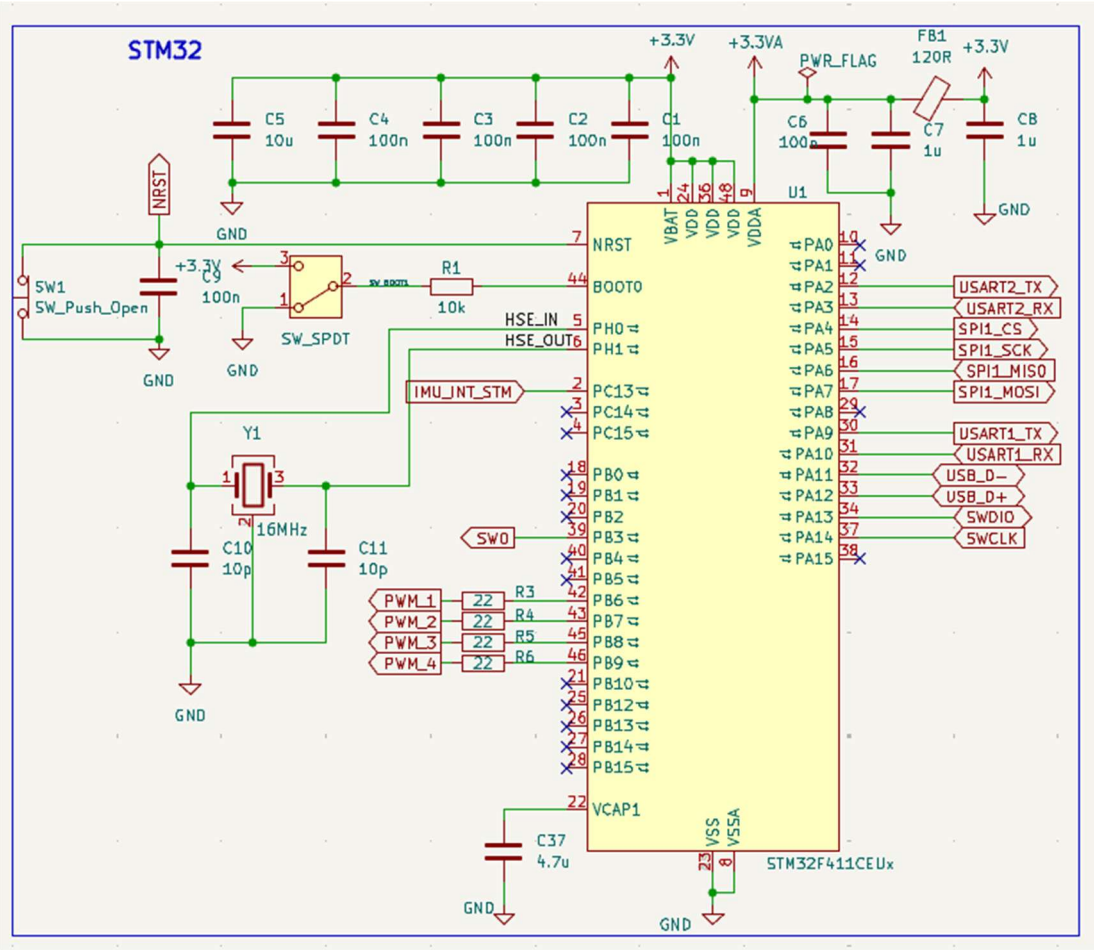
Block	Representative components / nets	Purpose
Main controller	U1 — STM32F411CEUx	Embedded control, peripheral communication and PWM generation.
Inertial sensing	U2 — ICM-20948	Accelerometer/gyroscope/magnetometer subsystem.
Primary power conversion	U5 — TPS54336ADDA; L1	Switching conversion associated with the +5 V rail.
3.3 V regulation	U3 — AMS1117-3.3	Generation of the +3.3 V rail.
1.8 V regulation	U4 — TLV75518PDBV	Generation of the +1.8 V rail.
Logic interface	U6–U10 — SN74AUP1T34DCK	Single-bit logic translation/buffering.
USB	J8 — Micro-B; USB_D+/USB_D−/USB_VBUS	USB connection to the controller.
Debug	J7 — 10-pin SWD/JTAG connector	Programming/debug access.
Battery input	J9 — XT60-Bat; F1; Q2	Battery connection and input protection/switching.
External control	PWM_1 … PWM_4; J3–J6	Control-signal connections.
Serial interfaces	USART1_RX/TX; USART2_RX/TX	External serial peripherals.

The PCB netlist also contains dedicated nets for SPI1 signals, including SPI1_SCK, SPI1_SCLK, SPI1_MOSI, SPI1_MISO, SPI1_SDI, SPI1_SDO, SPI1_CS and SPI1_NCS, as well as IMU interrupt nets.

3. Microcontroller Design

3.1 STM32F411

U1 is identified in the PCB as STM32F411CEUx and uses a QFN-48-1EP, 7 × 7 mm package with a 5.6 × 5.6 mm exposed pad. The schematic associates U1 with the STM32F411 datasheet.

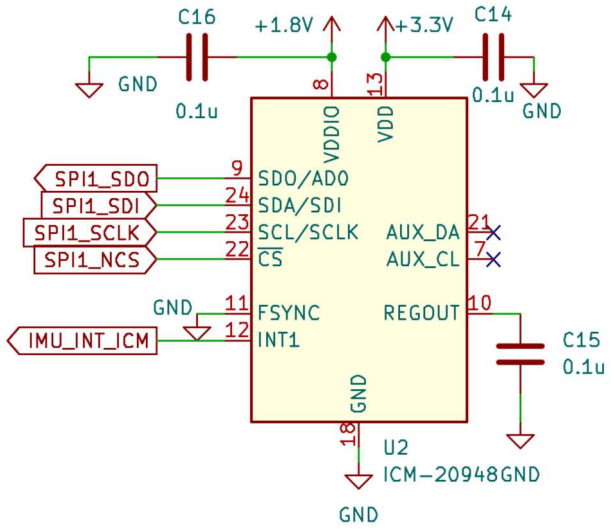


Design feature	Evidence in current PCB/schematic
MCU	U1 — STM32F411CEUx
Clock	Y1 — 16 MHz crystal; HSE_IN and HSE_OUT nets
Reset	NRST net
Boot	Net-(U1-BOOT0) and SW_BOOT1
Debug	SWDIO, SWCLK, SWO; J7
SPI	SPI1_* nets
Serial	USART1_RX/TX and USART2_RX/TX
PWM	PWM_1, PWM_2, PWM_3, PWM_4
IMU interrupt	IMU_INT_STM

The design therefore exposes the MCU resources required for a flight-controller architecture: a high-speed serial peripheral for the IMU, serial interfaces for external peripherals, PWM-labelled outputs for actuator interfaces, and SWD for development.

4. IMU Design

U2 is an ICM-20948 in a 24-pin 3 × 3 mm QFN package. The PCB identifies separate +1.8 V and +3.3 V power domains, and the netlist contains SPI1 signals and an IMU interrupt path.



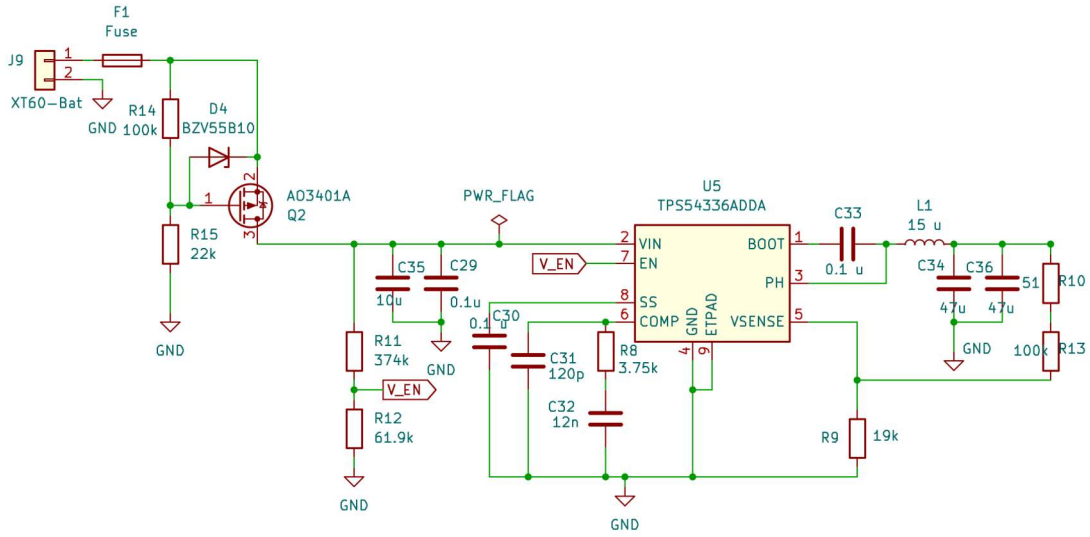
- U2 is the primary inertial-sensing component.
- SPI-related nets include SPI1_MOSI, SPI1_MISO, SPI1_SCK/SCLK, SPI1_CS and related IMU data/control nets.
- IMU_INT_ICM and IMU_INT_STM provide a dedicated interrupt path between the sensor and MCU domain.
- The PCB contains numerous 100nF local capacitors, consistent with a distributed decoupling strategy, although final placement quality must be verified visually before release.

5. Power Architecture

The design contains three explicitly named regulated rails: +5V, +3.3V and +1.8V. The power subsystem includes a switching regulator, a 3.3 V linear regulator, a 1.8 V LDO, input protection components, bulk capacitors and a ferrite bead.

Power element	Component	Current design value / part	Engineering role
Switching regulator	U5	TPS54336ADDA	Primary switching conversion stage.
Inductor	L1	15 u	Energy-storage element associated with the switching converter.
3.3 V regulator	U3	AMS1117-3.3	3.3 V linear regulation.
1.8 V regulator	U4	TLV75518PDBV	1.8 V regulation.
Ferrite bead	FB1	120R	High-frequency filtering / rail isolation.
Battery connector	J9	XT60-Bat	External battery connection.
Fuse	F1	Fuse; rating not specified in	Input protection.

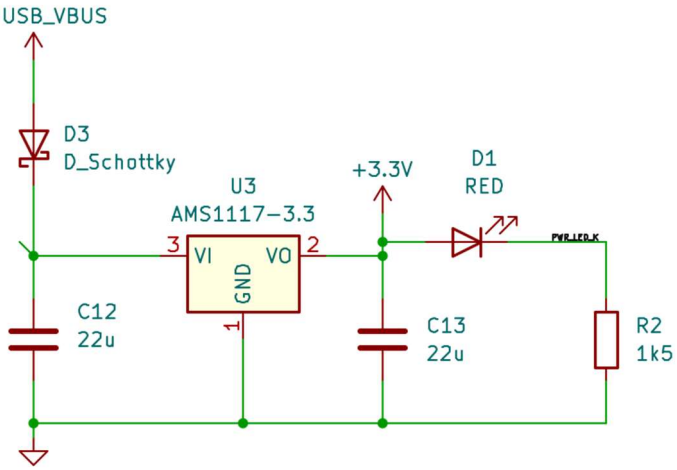
		PCB value	
MOSFET	Q2	AO3401A	Power-path switching/protection function; exact topology should be verified.
Bulk capacitors	C34, C36	47 uF	Bulk energy storage / ripple reduction.



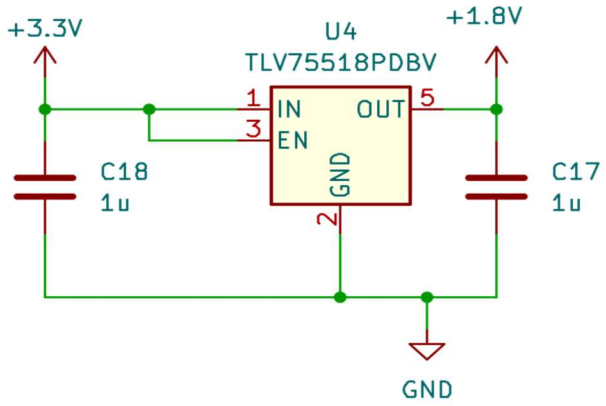
5.1 Switching regulator

U5 is a TPS54336ADDA with associated nets including Net-(U5-SS), Net-(U5-VSENSE), Net-(U5-PH), Net-(U5-COMP), Net-(U5-BOOT) and Net-(U5-SS). This indicates that the schematic implements the regulator's feedback, switching, compensation, soft-start and bootstrap-related circuitry rather than treating the converter as a black box.

5.2 3.3 V and 1.8 V rails

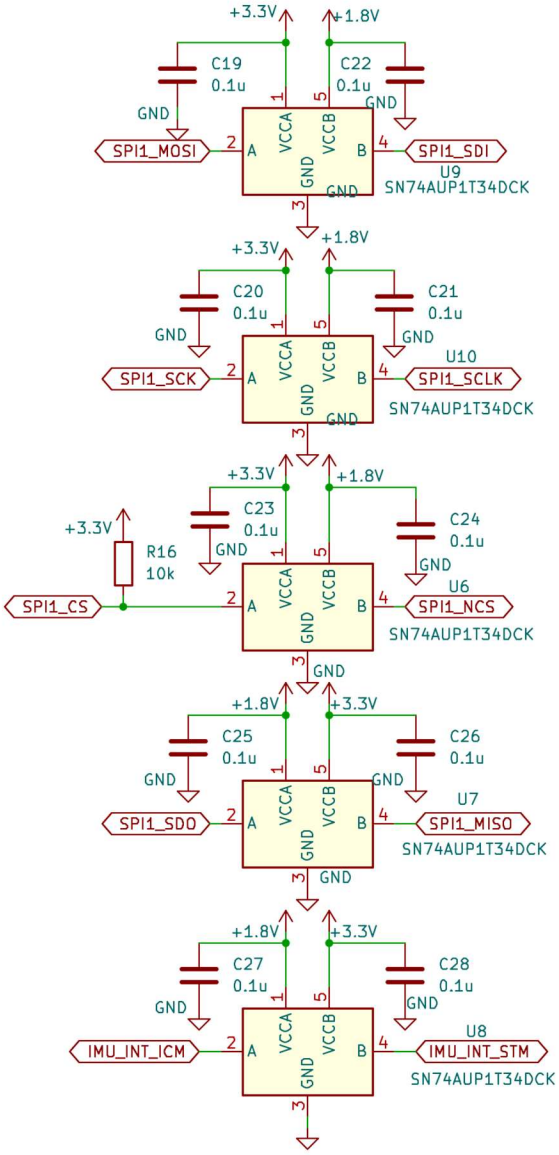


U3 is an AMS1117-3.3 and U4 is a TLV75518PDBV.



6. Logic-Level and Interface Translation

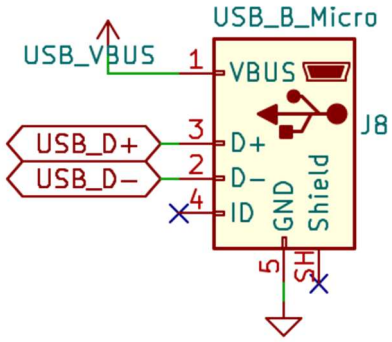
Five SN74AUP1T34DCK devices (U6–U10) are present. This is a single-bit logic buffer/translator family used in the design's logic-domain interfaces. The presence of the +1.8 V and +3.3 V rails makes voltage-domain compatibility a central consideration.



7. Communication and External Interfaces

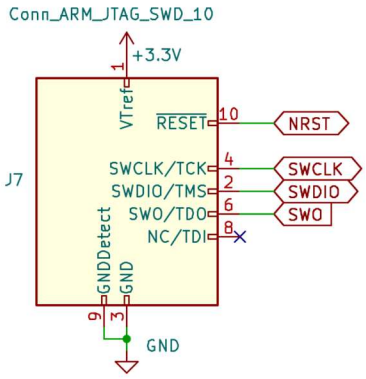
7.1 USB

J8 is a Micro-B USB connector. The PCB netlist explicitly defines USB_VBUS, USB_D− and USB_D+. The connector's ID pin and shield pads are shown as unconnected nets in the PCB.



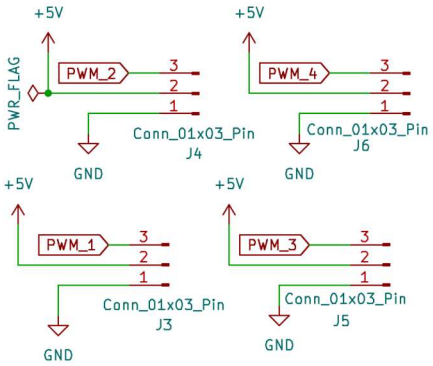
7.2 SWD / Debug

J7 is a 10-pin 1.27 mm SWD/JTAG-style connector. The board exposes +3.3V, GND, SWDIO, SWCLK, SWO and NRST. This provides a practical hardware-development interface for programming and debugging the STM32.



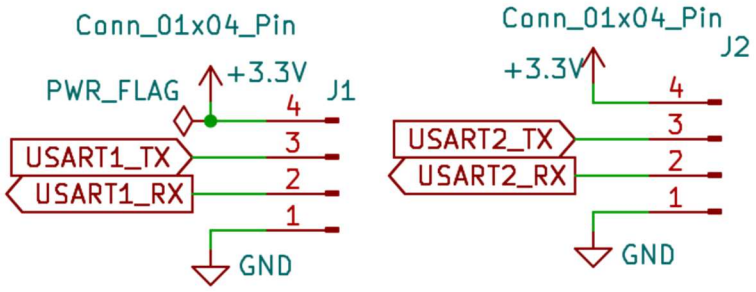
7.3 ESC / PWM interfaces

The PCB netlist contains PWM_1, PWM_2, PWM_3 and PWM_4, indicating four actuator-control outputs. The exact protocol and electrical meaning of these outputs are not established solely by the net names; firmware configuration must determine whether they are conventional PWM or another timing-based ESC protocol.



7.4 Serial interfaces

USART1_RX/TX and USART2_RX/TX are present as named nets. These provide two serial communication channels for external peripherals.



8. Clock, Reset and Boot Configuration

Y1 is a 16 MHz crystal using a 3.2 × 2.5 mm four-pin crystal footprint. The schematic/PCB exposes HSE_IN and HSE_OUT nets around the STM32. C10 and C11 are both 10 pF and are associated with the crystal network in the PCB inventory.

The board also includes a pushbutton switch SW1 and a boot switch SW_BOOT1. The netlist contains NRST and Net-(U1-BOOT0), indicating dedicated reset and boot-mode control.

9. PCB Construction and Mechanical Definition

9.1 Board dimensions

The current Edge.Cuts rectangle is 64.10 mm × 50.50 mm, giving an approximate board area of 3237.1 mm². The board outline is rectangular in the current PCB file.

9.2 Layer stack

Layer	Type	Copper thickness / dielectric
F.Cu	Outer copper	0.035 mm
Dielectric 1	FR4 prepreg	0.2064 mm; εr 4.5
In1.Cu	Inner copper	0.0152 mm
Dielectric 2	FR4 core	1.065 mm; εr 4.5
In2.Cu	Inner copper	0.0152 mm
Dielectric 3	FR4 prepreg	0.2064 mm; εr 4.5
B.Cu	Outer copper	0.035 mm

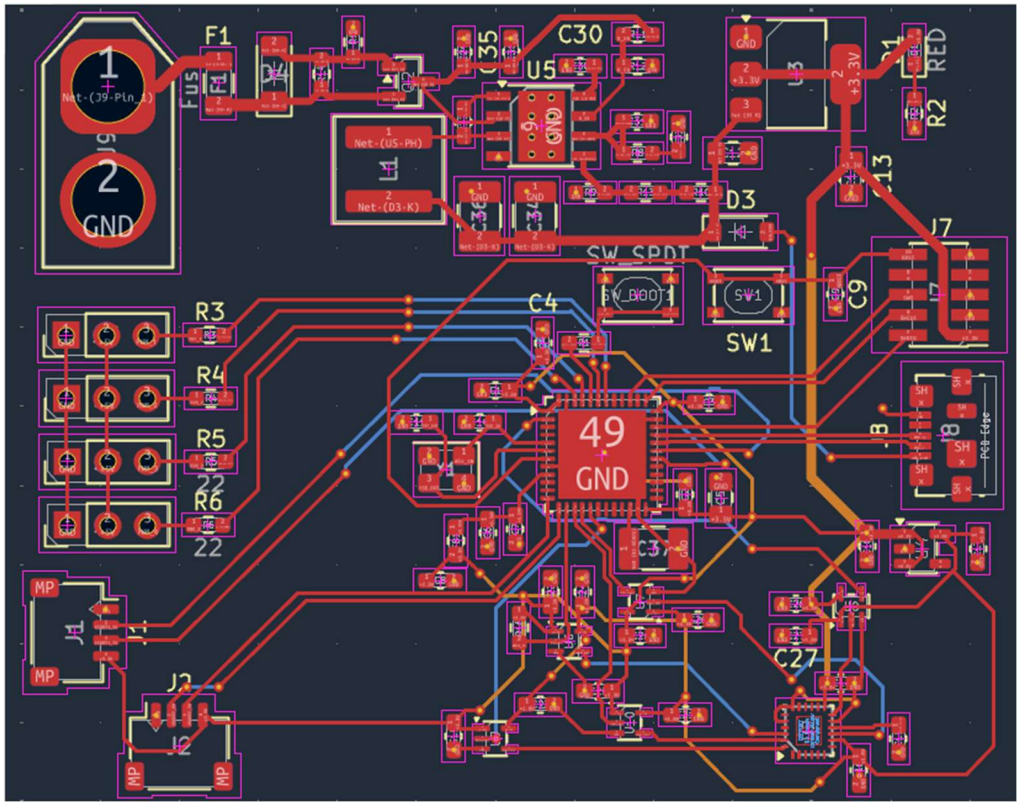
The stackup therefore provides two outer copper layers and two internal copper layers. The PCB file reports a nominal total thickness of 1.5982 mm. This is a strong basis for separating signal routing from reference-plane functions, but the final role assigned to each inner copper layer should be explicitly defined before manufacturing.

10. Component Placement and Routing

The current PCB contains 81 placed footprints: 37 capacitors, 15 resistors, 10 ICs, 9 connectors, 3 diodes, 2 switches, 2 fuse/protection elements, 1 inductor, 1 crystal and 1 MOSFET.

The design uses track widths of 0.20, 0.29, 0.40, 0.50, 0.60 and 0.80 mm in the current routed PCB. This indicates that routing width has been varied by net/function rather than using one global width.

Observed width	Observed use in PCB	Review
0.20 mm	Most routed segments	Suitable for many low-current signals, subject to manufacturer rules and current requirements.
0.29 mm	Small number of segments	Likely a specialized signal/power width; exact net purpose should be checked.
0.40 mm	Power/signal subset	Potentially appropriate for moderate-current paths; current and temperature rise should be verified.
0.50 mm	Power/signal subset	More robust routing for higher-current nets.
0.60 mm	Power subset	Useful for reducing resistance/voltage drop where space allows.
0.80 mm	Small number of segments	Likely intended for higher-current power paths; verify against actual load.



10.1 Placement priorities

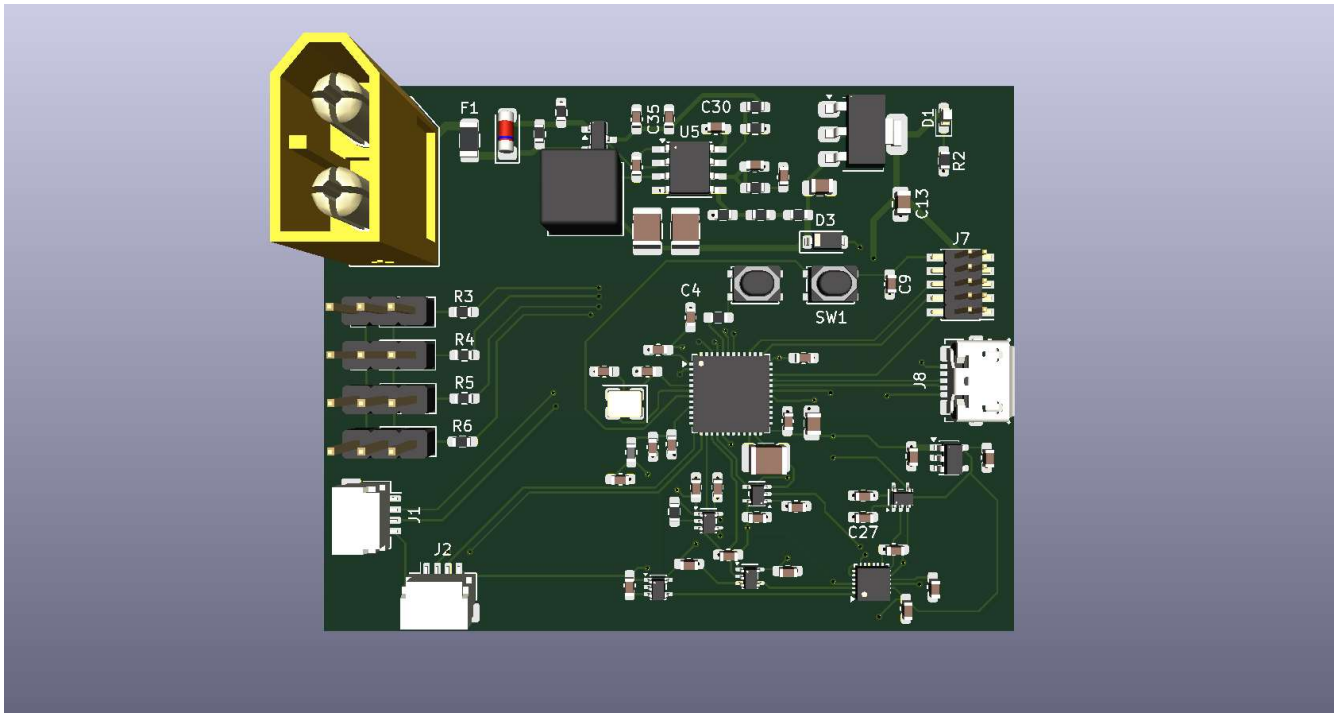
- Keep the IMU away from strong switching-current loops and mechanically undesirable vibration sources.
- Place regulator input/output capacitors close to the relevant regulator pins.
- Keep the switching regulator's high di/dt loop compact.
- Maintain a continuous reference plane under sensitive digital interfaces where practical.
- Keep USB D+/D− routing short, symmetric and referenced to a suitable plane.
- Keep high-current battery paths physically separated from sensitive sensor and clock circuitry.

11. Signal Integrity and Power Integrity

11.1 Signal integrity

The four-layer construction provides a useful foundation for controlled return paths. The most sensitive interfaces in this design are expected to be USB, SPI to the IMU, the 16 MHz clock network, and fast MCU-to-buffer signals.

- Avoid unnecessary layer transitions on fast signals.
- Provide a continuous reference plane underneath high-speed traces.
- Avoid routing fast signals through large plane splits or voids.
- Keep SPI clock routing controlled and avoid long parallel runs beside unrelated high-speed signals.
- Treat USB D+/D− as a differential pair and verify geometry against the final stackup.



11.2 Crosstalk

Crosstalk risk depends on trace spacing, parallel run length, edge rate, dielectric geometry and return path. The PCB contains multiple 0.20 mm signal traces, so spacing should be checked locally in the finished routing rather than assumed from the global design.

11.3 Power integrity

The board contains many 100 nF capacitors and several 1 µF, 10 µF, 22 µF and 47 µF capacitors. This provides both local high-frequency bypassing and larger energy-storage elements. Final assessment should verify that every regulator and IC has the capacitor type, value, voltage rating and placement required by its datasheet.

12. Component Selection Rationale

This section explains why each power, protection and interface component is on the board, why its value was chosen, and what the alternatives were.

12.1 Reverse-polarity protection: Q2 AO3401A P-channel MOSFET

What: Q2 is a P-channel MOSFET in series with the battery positive line, after fuse F1. R15 (22 kΩ) pulls the gate towards GND, R14 (100 kΩ) sits between gate and source, and zener D4 (BZV55B10, 10 V) clamps the gate-source voltage.

Why reverse-polarity protection is needed: an XT60 connector is keyed, but pigtails, adapters and bench supplies are not. A reversed 3S/4S pack applies −12.6 V to −16.8 V to the TPS54336A, whose absolute minimum on VIN is −0.3 V. Without protection the regulator, and everything powered from it, would be destroyed.

Why a MOSFET instead of a series diode: a series Schottky diode also blocks reverse current, but it drops about 0.3–0.5 V permanently. At the estimated board current of 0.13 A (Section 13.5) the diode would waste about 52 mW and reduce input headroom. Q2, fully enhanced, has RDS(on) below 50 mΩ, so it dissipates under 1 mW and drops about 6.5 mV.

Option	Loss at 0.13 A	Voltage drop	Comment
Series Schottky diode	≈ 52 mW	0.3–0.5 V	Simplest; permanent drop and heat
P-MOSFET, high side (this design)	< 1 mW	≈ 6.5 mV	Ground path unbroken; needs gate clamp
N-MOSFET, low side	< 1 mW	≈ mV	Opens the battery-negative path

Why P-channel, high side: a P-MOSFET in the positive line turns on when its gate is pulled below its source, so a resistor to GND is enough; no charge pump or bootstrap is needed. A low-side N-MOSFET would be cheaper and lower-resistance, but it places a switch in the battery negative. On a drone, the board ground is also tied to the ESC signal grounds, so a low-side switch could let return current find an alternative path through the signal wiring. Keeping the ground path continuous is the safer choice here.

Why AO3401A: its −30 V VDS rating covers a full 4S pack (16.8 V) with about 1.8× margin, its −4 A continuous rating is far above the board's current, and RDS(on) is specified at gate drives of −10 V, −4.5 V and −2.5 V (< 50, < 60 and < 85 mΩ). Its gate-source rating is only ±12 V, which is why a gate clamp is needed (Section 12.2).

12.2 Gate network: R14, R15 and zener D4

R15 (22 kΩ, gate to GND) provides the turn-on drive. **R14 (100 kΩ, gate to source)** forms a divider with R15 and discharges the gate when the battery is removed or reversed, so Q2 cannot stay on. **D4** limits the gate-source voltage to 10 V.

Why the clamp is needed: the divider alone gives VGS = −0.82 × Vbat. At a full 3S pack (12.6 V) that is −10.3 V, just inside the ±12 V rating, but at a full 4S pack (16.8 V) it would be −13.8 V, beyond the rating. The DJI F450 frame is specified for 3S–4S packs, so the clamp keeps the circuit safe on 4S and during supply transients. Numbers are in Section 13.4.

Why these values: the large resistances keep the standing current to about 0.1 mA at 12.6 V (1.3 mW), which is negligible next to the board load. The 100 kΩ : 22 kΩ ratio puts the gate drive at −4.5 V or beyond once the battery exceeds 5.5 V, so Q2 is fully enhanced over the whole 3S and 4S range. The trade-off is a slow turn-off (gate capacitance discharging through 100 kΩ), which is acceptable because this is a static protection switch, not a switching element.

12.3 Input fuse F1

What: a 1206 SMD fuse between the XT60 positive pin and Q2.

Why: a LiPo can deliver very large fault currents. F1 protects the battery leads and board copper if a fault downstream (a failed regulator, a solder bridge) shorts the input. It does not protect the semiconductors, which fail faster than any fuse opens.

12.4 Primary regulator: TPS54336A synchronous buck converter (U5)

Why a switching regulator: generating 3.3 V directly from the battery with a linear regulator at a 0.2 A load would dissipate (12.6 − 3.3) × 0.2 = 1.86 W at a full 3S pack and 2.7 W at 4S. That is far too much for a SOT-223 package. The buck stage converts the battery to 5 V at an estimated IC loss of about 19 mW (Section 13.11), so the linear regulators that follow only drop 1.7 V and 1.5 V.

Why TPS54336A: its 4.5–28 V input range covers 3S and 4S with transient margin; the synchronous MOSFETs (128 mΩ high side, 84 mΩ low side) are integrated; the 340 kHz switching frequency is fixed; soft-start and loop compensation are set externally; and it enters pulse-skipping at light load, which keeps efficiency up at the small currents this board draws. Its 3 A capability is much more than the board needs, leaving headroom for a receiver, GPS or a powered ESC rail.

Function	TI reference design	This board
Inductor	15 µH (Coilcraft XAL6060-153MEB)	L1 15 µH, XAL6060 footprint
Output capacitors	2 × 47 µF, 10 V X5R	C34, C36: 2 × 47 µF, 1210 (rating not specified)
Feedback divider	100 kΩ / 19.1 kΩ	R13 100 kΩ / R9 19 kΩ
Loop-injection resistor	51.1 Ω	R10 51 Ω
Compensation	3.74 kΩ, 12 nF, 120 pF	R8 3.75 kΩ, C32 12 nF, C31 120 pF
Bootstrap capacitor	0.1 µF	C33 0.1 µF
Soft-start capacitor	10 nF (≈ 3.5 ms)	C30 100 nF (≈ 35 ms)
Input capacitors	10 µF, 35 V X7R + 0.1 µF	C35 10 µF 0603 (rating not specified) + C29 0.1 µF

Trade-offs: the part is over-rated for this load, and at light load it pulse-skips, which raises output ripple compared with continuous operation. A smaller 1 A converter at a higher frequency would save board area. The benefit of this choice is a well-documented, measured reference design and a 5 V rail with large current headroom.

12.5 Buck inductor L1 (15 µH, Coilcraft XAL6060 footprint)

Why 15 µH: it is the value from the TI reference design. At this board's light load the ripple current (0.44–0.69 A peak-to-peak, Section 13.7) is larger than the average load current, so the converter runs in pulse-skipping or discontinuous mode. This is expected behaviour for TPS54336A below about 0.5 A peak switch current and is not a fault.

Why a shielded, molded inductor: molded inductors such as the XAL family have soft saturation and contain their magnetic field. This matters on a board where a magnetometer (inside the ICM-20948) sits on the same PCB.

12.6 USB power OR-ing: Schottky diode D3

What: D3 connects USB VBUS (anode) to the 5 V buck output node (cathode). **Why:** it lets the board run from USB alone for programming and configuration, without a battery. It also stops the 5.01 V buck output from driving current back into the USB host when the battery is connected. That back-feed would push current into a PC port or charger that is not designed to accept it.

Why a Schottky diode: its forward drop (about 0.3–0.5 V, ASSUMPTION until the MPN is known) is roughly half that of a silicon diode. That matters because the AMS1117 needs more than 1 V of headroom (Section 13.15). **Trade-offs:** a Schottky diode has higher reverse leakage, which back-feeds a few microamps into VBUS when battery-powered, and it still costs some headroom. An ideal-diode controller or USB load switch would remove both drawbacks at higher cost and complexity.

12.7 3.3 V regulator: AMS1117-3.3 (U3)

Why a linear regulator after the buck: a linear stage rejects part of the 340 kHz switching ripple before it reaches the MCU, the IMU supply and the USB transceiver. It costs only $(5.01 - 3.3) \times I$, which is 0.34 W at a 0.2 A budget (Section 13.13), and a SOT-223 tab can spread that heat into the copper.

Trade-offs: the AMS1117 family has a high dropout of about 1 V or more. That is irrelevant when running from the 5 V buck, but it is tight when the board is powered from USB through D3 (Section 13.15). A low-dropout CMOS regulator would remove this limitation.

12.8 1.8 V regulator: TLV75518PDBV (U4)

Why a 1.8 V rail exists: the ICM-20948 accepts 1.71–3.6 V on VDD but only **1.71–1.95 V on VDDIO**. Its digital I/O therefore cannot be connected directly to the 3.3 V STM32. The board powers the IMU core (VDD) from 3.3 V and its I/O (VDDIO) from 1.8 V.

Why the MCU is not run at 1.8 V instead: the STM32F411 can operate down to 1.7 V, but its USB full-speed PHY is specified for VDD ≥ 3.0 V (characteristics degrade from 2.7 V to 3.0 V), and the external receiver, GPS and ESC interfaces expect 3.3 V logic. A 1.8 V island around the IMU is therefore the smaller change.

Why TLV75518 from 3.3 V: its input range (1.45–5.5 V) accepts 3.3 V, and its dissipation from 3.3 V is only (3.3 − 1.8) × I, about 15 mW at 10 mA. It is stable with a 1 µF ceramic output capacitor (C17 output, C18 input), and its active output discharge gives a defined state at power-up. EN is tied to IN, so the rail is always on when 3.3 V is present.

12.9 Logic-level translation: five SN74AUP1T34DCK (U6–U10)

Why translation is needed: the ICM-20948 inputs switch at 0.7 × VDDIO (about 1.26 V) and its outputs swing only to 1.8 V, while the STM32 drives 3.3 V. Driving 3.3 V into a 1.8 V-referenced input exceeds what VDDIO is specified for, and a 1.8 V output is not a guaranteed logic high for every 3.3 V input.

Why single-bit, unidirectional translators: every SPI and interrupt signal has a fixed direction, so a dedicated buffer per signal gives a defined push-pull drive in each direction. Auto-direction translators sense which side is driving, and they can misbehave with pull-up resistors or heavy loads; with fixed directions that flexibility is not needed. The SN74AUP1T34 also has Ioff partial power-down and puts both ports into high impedance if either supply is at GND, which protects the IMU while rails ramp.

Ref.	A side (input)	B side (output)	Direction	Signal
U6	SPI1_CS (3.3 V)	SPI1_NCS (1.8 V)	MCU → IMU	Chip select, PA4
U10	SPI1_SCK (3.3 V)	SPI1_SCLK (1.8 V)	MCU → IMU	SPI clock, PA5
U9	SPI1_MOSI (3.3 V)	SPI1_SDI (1.8 V)	MCU → IMU	Data to IMU, PA7
U7	SPI1_SDO (1.8 V)	SPI1_MISO (3.3 V)	IMU → MCU	Data from IMU, PA6
U8	IMU_INT_ICM (1.8 V)	IMU_INT_STM (3.3 V)	IMU → MCU	Data-ready interrupt, PC13

Trade-off: each translator adds a propagation delay of about 5 ns on every SPI edge, which limits the usable SPI clock.

12.10 Analog supply filter: FB1 (120 Ω ferrite bead), C6, C7

What: FB1 feeds +3.3VA (STM32 VDDA, pin 9) from +3.3V, with C6 (100 nF) and C7 (1 µF) on the filtered side. **Why:** VDDA powers the ADC, the reset circuitry, the internal RC oscillators and the PLL. Digital switching noise on +3.3V would add jitter to the PLL and error to ADC readings. The 100 nF + 1 µF pair follows ST's recommended decoupling for VDDA.

Why a ferrite bead and not a resistor: a ferrite is nearly lossless at DC but becomes resistive at high frequency, where digital noise lives. ST also requires VDDA to stay at the same potential as VDD (within 300 mV during power-up). A ferrite keeps the DC drop to around a millivolt, whereas a resistor large enough to filter would drop too much.

Why 120 Ω: a ferrite's rating is its impedance at a stated test frequency, usually 100 MHz. 120 Ω is a common middle choice: high enough to dominate C7's impedance in the tens-to-hundreds of MHz noise band, with a DC resistance low enough to be negligible. The filter attenuation, DC drop and resonance are estimated in Section 13.18.

12.11 ESC signal series resistors R3–R6 (22 Ω)

What: R3–R6 sit in series between PB6–PB9 (TIM4 CH1–CH4) and the PWM_1–PWM_4 header pins. **Why:** the ESC lines leave the board through wires several centimetres long. A small series resistor damps ringing caused by the fast GPIO edge driving a long, unterminated line. It also slows the edge, which reduces radiated noise, and it limits the current that a voltage spike on the wire can push into the MCU pin's protection structure.

Why 22 Ω: the RC time constant with an assumed 100 pF of wire and ESC input capacitance is 2.2 ns. That is negligible against a DShot600 bit time of 1.67 µs, so the protocol timing is unaffected.

12.12 Reset and boot network: C9, SW1, R1, SW_BOOT1

NRST: C9 (100 nF) from NRST to GND, together with the STM32's internal pull-up, filters noise on the reset pin; this is ST's recommended NRST protection. SW1 (PTS810, normally open) pulls NRST low to reset.

BOOT0: BOOT0 selects the boot source. Low runs the user flash; high runs the ST system bootloader (USB DFU, UART). R1 (10 kΩ) connects BOOT0 to SW_BOOT1.

12.13 Clock, power indicator and decoupling

- **Y1 16 MHz crystal with C10/C11 (10 pF):** a crystal is used instead of the internal 16 MHz RC oscillator because USB full speed needs an accurate 48 MHz clock derived from HSE through the PLL. The 10 pF load capacitors set the crystal's load capacitance; Section 13.17 shows that the crystal must be specified for about 7–10 pF.
- **D1 + R2 (1.5 kΩ) power LED:** indicates that the 3.3 V rail is up. The current is about 0.9 mA (Section 13.16), visible indoors while negligible in the power budget.
- **MCU decoupling:** C1–C4 (100 nF) at the VDD pins, plus C5 (10 µF) and C8 (1 µF) as bulk on +3.3V, follow ST's scheme of 100 nF per VDD pin plus a ≥ 4.7 µF reservoir. C37 (4.7 µF) on VCAP_1 meets the datasheet requirement for a single VCAP pin: 4.7 µF with ESR below 1 Ω.
- **IMU decoupling:** C14 (VDD), C16 (VDDIO) and C15 (REGOUT), each 0.1 µF, match the three capacitors in the ICM-20948 reference circuit. The IMU footprint correctly has no exposed pad, as the datasheet says the die pad must not be soldered.

13. Engineering Calculations

Each calculation shows the formula, the substitution with this board's values, and the result, followed by what the result means for the hardware.

13.1 Buck output voltage (feedback divider R13 + R10, R9)

The TPS54336A regulates VSENSE to its 0.8 V reference. The upper leg is R13 (100 kΩ) in series with the 51 Ω loop-injection resistor R10; the lower leg is R9 (19 kΩ).

$$Vout = Vref * (1 + \frac{R_top}{R_bottom})$$

$$Vout = 0.8 * (1 + \frac{100\ k\Omega + 51\Omega}{19.1\ k\Omega})$$

$$Vout = 4.991V$$

Tolerance (1 % resistors, Vref 0.788–0.812 V over temperature):

Vout,max = 0.812 × (1 + 101.05 kΩ / 18.81 kΩ) = 5.17 V

Vout,min = 0.788 × (1 + 99.05 kΩ / 19.19 kΩ) = 4.86 V

Meaning: the rail is 5.0 V ±3.2 % worst case, which is acceptable for the AMS1117 input and for 5 V peripherals. **Why 51 Ω in the top leg:** inserting a small resistor at the top of the divider gives a place to inject a test signal and measure the loop gain and phase with a network analyser, without changing the DC set-point (it adds only 0.05 %).

13.2 Feedback divider current

$$I_{div} = \frac{V_{out}}{R_{top} + R_{bottom}}$$

$$I_{div} = \frac{5.013V}{100.05\ k\Omega + 19\ k\Omega} = 42\ \text{uA}$$

Meaning: TI suggests an upper resistor of about 100 kΩ. Larger values reduce the divider's standing loss, which matters for light-load efficiency; values that are too large make VSENSE sensitive to noise and to the pin's input current. 42 µA is a negligible load (0.2 mW).

13.3 Undervoltage lockout divider (R11, R12)

R11 (374 kΩ) from VIN to EN and R12 (61.9 kΩ) from EN to GND hold the converter off until the battery is high enough. The EN pin also sources a pull-up current Ip = 1.15 µA, which rises by Ih = 3.3 µA once the device is enabled. That step creates hysteresis. The datasheet thresholds are 1.21 V rising and 1.17 V falling.

Start:

$$V_{start} = R11 * \frac{V_{en},rise}{R12 - I_p} + V_{EN,rise}$$

$$V_{start} = 374\text{k}\Omega * \frac{1.21V}{61.9\ \text{k}\Omega - 1.15\ uA} + 1.21\ V = 8.09V$$

Stop:

$$V_{start} = R11 * \frac{V_{en},fall}{R12 - (I_p + I_h)} + V_{EN,fall}$$

$$V_{start} = 374\text{k}\Omega * \frac{1.17V}{61.9\ \text{k}\Omega - 4.45\ uA} + 1.17\ V = 6.57V$$

Hysteresis = 8.09 V − 6.57 V = 1.52 V

EN voltage at 16.8 V input (4S full) ≈ 2.62 V (absolute maximum 6 V)

Meaning: the converter starts at 8.09 V and stops at 6.57 V. Both are below a depleted 3S pack (about 9 V at 3.0 V per cell), so the flight controller stays powered across the whole discharge. The 1.52 V hysteresis exceeds TI's recommended minimum of 500 mV, which prevents on/off chatter when the pack sags under motor load. The EN pin stays well below its 6 V limit even on 4S.

13.4 Q2 gate drive and D4 clamp

$$V_{GS} = -V_{bat} * \frac{R14}{R14 + R15}$$

$$V_{GS} = -V_{bat} * \frac{100\text{k}\Omega}{122\ \text{k}\Omega} = -0.82 * V_{bat}$$

Full enhancement (V_GS ≤ −4.5 V): Vbat ≥ 4.5 V / 0.820 = 5.49 V

Battery state	Vbat	Unclamped V_GS	Actual V_GS
3S empty	9.0 V	−7.4 V	−7.4 V
3S nominal	11.1 V	−9.1 V	−9.1 V
3S full	12.6 V	−10.3 V	−10 V (clamped)
4S full	16.8 V	−13.8 V	−10 V (clamped)

```
Zener current at 4S full (V_GS clamped at -10 V):
  I_R15 = (16.8 V - 10 V) / 22 kΩ  = 309 µA
  I_R14 = 10 V / 100 kΩ            = 100 µA
  I_Z   = 309 µA - 100 µA          = 209 µA
  P_Z   = 10 V × 209 µA            = 2.1 mW   (BZV55B10 rating 500 mW)
Standing divider current at 12.6 V: 12.6 V / 122 kΩ = 103 µA
```

Meaning: Q2 is fully enhanced (RDS(on) < 60 mΩ at −4.5 V, < 50 mΩ at −10 V) everywhere in the 3S and 4S range. Without D4, a 4S pack would exceed the ±12 V gate rating; with D4 the gate never sees more than 10 V, and the zener runs at 0.4 % of its power rating.

13.5 Battery current and Q2 conduction loss

```
Formula:       I_bat = (Vout × Iout) / (η × Vbat)
Substitution:  I_bat = (5.01 V × 0.2 A) / (0.85 × 9 V)
Result:        I_bat = 0.131 A

Q2 loss:           P = I² × RDS(on) = 0.131² × 0.05 Ω = 0.86 mW
Schottky instead:  P = Vf × I = 0.4 V × 0.131 A        = 52 mW
```

Meaning: the MOSFET wastes about 60 times less power than a diode would, and its 6.5 mV drop does not affect the UVLO thresholds.

13.6 Soft-start time and inrush current (C30)

```
Formula:       t_SS = C_SS × Vref / I_SS
Substitution:  t_SS = 100 nF × 0.8 V / 2.3 µA
Result:        t_SS = 34.8 ms

Output-capacitor charging current: I = C_out × Vout / t_SS
                                   = 94 µF × 5.01 V / 34.8 ms = 13.5 mA
```

Meaning: the 5 V rail rises in about 35 ms, ten times slower than TI's 10 nF example (3.5 ms). The inrush current needed to charge the output capacitors is only 13.5 mA, so the soft-start creates no stress for F1, Q2 or a USB host. The slower ramp is harmless; it delays MCU start-up by a few tens of milliseconds. Using 10 nF is optional.

13.7 Inductor ripple and peak current

```
Formula:       ΔI_L = Vout × (Vin − Vout) / (Vin × L × f_sw)
Substitution:  ΔI_L = 5.01 V × (12.6 V − 5.01 V) / (12.6 V × 15 µH × 340 kHz)
Result:        ΔI_L = 0.59 A peak-to-peak (at 12.6 V)

Peak current:  I_pk = I_out + ΔI_L / 2
               I_pk = 0.25 A + 0.69 A / 2 = 0.59 A  (16.8 V worst case)
```

Vin	Duty cycle	On-time	ΔI_L (p-p)	Comment
9.0 V (3S empty)	0.557	1.64 µs	0.44 A	Lowest ripple
12.6 V (3S full)	0.398	1.17 µs	0.59 A	Nominal design point
16.8 V (4S full)	0.298	0.88 µs	0.69 A	Highest ripple

Meaning: the shortest on-time (0.88 µs) is far above the TPS54336A minimum controllable on-time (94 ns typ, 145 ns max), so the converter can regulate across the full input range. The peak current of about 0.6 A is roughly one tenth of the XAL6060-153 saturation current (5.8 A). Because the ripple exceeds twice the load current, the converter operates in pulse-skipping or discontinuous mode at this board's load, which is normal for this device.

13.8 Output capacitors C34, C36 (2 × 47 µF)

ASSUMPTION: load step 0.2 A, allowed transient deviation 5 % (0.25 V), allowed ripple 30 mV (the same criteria as TI's example, scaled to this load).

```
Transient:  C_out > 2 × ΔI_out / (f_sw × ΔV_out)
                  = 2 × 0.2 A / (340 kHz × 0.25 V) = 4.7 µF
Ripple:     C_out > ΔI_L / (8 × f_sw × V_ripple)
                  = 0.59 A / (8 × 340 kHz × 30 mV) = 7.2 µF

Expected ripple (capacitive):  ΔV = ΔI_L / (8 × f_sw × C_eff)
  C_eff = 94 µF (nominal):           ΔV = 2.3 mV
  C_eff = 47 µF (50 % DC-bias loss): ΔV = 4.6 mV
ESR term (3 mΩ assumed, as TI):     ΔV = 0.59 A × 3 mΩ = 1.8 mV
```

Meaning: the board needs only about 7 µF to meet the ripple and transient targets. The 2 × 47 µF are larger because they set the converter's output pole, and the compensation network (Section 13.10) was tuned by TI for this capacitance. The capacitors therefore must keep enough effective capacitance at 5 V bias; ceramic capacitors lose a large fraction of their nominal value under DC bias, especially in small packages. Use X5R or X7R parts rated 10 V or higher in 1210, as on the board, and check the manufacturer's DC-bias curve.

13.9 Input capacitor C35 (10 µF) and C29 (100 nF)

```
Input ripple:  ΔV_in ≈ I_out × 0.25 / (C_in × f_sw)
  C_eff = 10 µF:                        ΔV_in = 18 mV
  C_eff = 3 µF (0603 at 12 V, ASSUMED): ΔV_in = 61 mV
RMS current:   I_Cin,rms ≈ I_out / 2 = 0.125 A
```

Meaning: the ripple is small either way. The real constraints are the **voltage rating** (above 16.8 V for 4S, with transient margin) and DC-bias derating. A 10 µF capacitor in 0603 loses most of its capacitance at 12–17 V. C29 (100 nF) handles the highest-frequency edge currents and must sit closest to the VIN and GND pins, as TI's layout guide requires.

13.10 Loop compensation (R8, C32, C31)

TPS54336A uses peak-current-mode control with a transconductance error amplifier (gm_ea = 1300 µA/V) and a power-stage transconductance gm_ps = 8 A/V. R8 and C32 form a Type II compensator; C31 adds a high-frequency pole that filters switching noise.

```
Compensation zero: f_z = 1 / (2π × R8 × C32) = 1 / (2π × 3.75 kΩ × 12 nF) = 3.54 kHz
High-freq. pole:   f_p = 1 / (2π × R8 × C31) = 1 / (2π × 3.75 kΩ × 120 pF) = 354 kHz
Modulator pole:      f_pm = I_out / (2π × Vout × C_out)
                          = 0.1–0.25 A / (2π × 5.01 V × 94 µF)              = 34–84 Hz

Crossover (simple model): f_c = R8 × gm_ea × Vref × gm_ps / (2π × Vout × C_out)
  C_out = 94 µF: f_c = 3.75 kΩ × 1.3 mA/V × 0.8 V × 8 A/V / (2π × 5.01 V × 94 µF)
                     = 10.5 kHz
  C_out = 47 µF: f_c = 21.1 kHz
Upper limit (f_sw / 10): 34 kHz
```

Meaning: the zero (3.5 kHz) lies below crossover, which provides phase boost; the high-frequency pole (354 kHz) lies near the switching frequency, which attenuates switching ripple in the control loop. The estimated crossover stays below one tenth of the switching frequency for any realistic effective output capacitance. TI chose these values from a measured power-stage response (31.6 kHz target crossover). The simple model ignores slope compensation, so it is only a first-order check.

13.11 Buck converter IC dissipation

ASSUMPTION: 0.25 A load at 12.6 V. TI's formulas assume continuous conduction; at this load the device pulse-skips, so treat this as an order-of-magnitude estimate.

```
Conduction:  P_con = I² × R_HS × Vout / Vin = 0.25² × 0.128 × 5.01 / 12.6 = 3.2 mW
Switching:   P_sw  = 0.5×10⁻⁹ × Vin² × I × f_sw
                   = 0.5×10⁻⁹ × 12.6² × 0.25 × 340 kHz = 6.7 mW
Gate charge: P_g   = 22.8×10⁻⁹ × f_sw = 22.8×10⁻⁹ × 340 kHz = 7.8 mW
Quiescent:   P_q   = 0.11×10⁻³ × Vin  = 0.11×10⁻³ × 12.6 V  = 1.4 mW
Total:       P_tot ≈ 19 mW    →  ΔT_J = 42.1 °C/W × 19 mW ≈ 0.8 °C
Inductor copper loss: I² × DCR = 0.25² × 43.8 mΩ = 2.7 mW
```

Meaning: thermally, the buck converter is not a concern at this load. Its PowerPAD still needs its thermal vias to the ground plane, because the pad is also the device's ground connection.

13.12 Why not a single linear regulator

```
P = (Vin − Vout) × I
  3S full: (12.6 V − 3.3 V) × 0.2 A = 1.86 W
  4S full: (16.8 V − 3.3 V) × 0.2 A = 2.70 W
```

Meaning: a SOT-223 package on a small board cannot dissipate 1.9–2.7 W without exceeding its junction temperature. This is the main justification for the buck-plus-LDO architecture.

13.13 3.3 V load budget and AMS1117 dissipation

Load on +3.3V	Current	Source
STM32F411 core at 100 MHz	≈ 10 mA	FACT: datasheet run current ≈ 100 µA/MHz with peripherals off; peripherals add more
ICM-20948, 9-axis low-noise	3.11 mA	FACT: datasheet
SN74AUP1T34 × 5	< 0.03 mA	FACT: ≤ 5.2 µA each (static)
Power LED D1	0.87 mA	Calculated, Section 13.16
TLV75518 → 1.8 V loads	< 5 mA	IMU VDDIO + translator B sides (ANALYSIS)
GPS and receiver on J1/J2 pin 4	unknown	Depends on modules not yet specified
Design budget	**200 mA**	ASSUMPTION covering a GPS and a receiver

```
Formula:       P_U3 = (V_in − V_out) × I_load
Substitution:  P_U3 = (5.01 V − 3.3 V) × 0.2 A
Result:        P_U3 = 0.34 W

Maximum allowable θJA for T_J ≤ 125 °C at T_A = 50 °C (ASSUMED ambient):
  θJA,max = (125 °C − 50 °C) / 0.34 W = 219 °C/W
```

Meaning: a SOT-223 regulator on copper is normally specified well below 219 °C/W, so 0.34 W is acceptable. The exact θJA depends on the AMS1117 vendor and on the copper area attached to the tab (pin 2 = VOUT), so it must be confirmed from the chosen part's datasheet.

13.14 1.8 V rail (TLV75518)

```
P_U4 = (3.3 V − 1.8 V) × 10 mA = 15 mW
Headroom = 3.3 V − 1.8 V = 1.5 V   (dropout ≤ 0.40 V at 500 mA, 1.8 V output)
```

Meaning: the 1.8 V regulator has ample headroom and negligible dissipation. C17 and C18 (1 µF each) meet the requirement of at least 1 µF nominal and at least 0.47 µF effective on both input and output.

13.15 USB-powered headroom (D3 → AMS1117)

ASSUMPTION: VBUS 4.75–5.25 V (USB 2.0 port range); D3 forward drop 0.3–0.5 V until the MPN is known; AMS1117-class dropout of about 1.1–1.3 V at moderate load (vendor-dependent).

VBUS	D3 Vf	AMS1117 input	Headroom above 3.3 V
4.75 V	0.5 V	4.25 V	0.95 V
4.75 V	0.3 V	4.45 V	1.15 V
5.25 V	0.5 V	4.75 V	1.45 V
5.25 V	0.3 V	4.95 V	1.65 V

```
D3 dissipation at the 500 mA USB limit: P = 0.45 V × 0.5 A ≈ 0.23 W
```

13.16 Power LED current (R2)

ASSUMPTION: red LED forward voltage 1.8–2.2 V (LED MPN not specified).

```
Formula:       I_LED = (V_3V3 − V_F) / R2
Substitution:  I_LED = (3.3 V − 2.0 V) / 1.5 kΩ
Result:        I_LED = 0.87 mA   (range 0.73–1.0 mA for V_F 2.2–1.8 V)
```

Meaning: about 1 mA is enough for a modern 0603 LED to be visible indoors while costing only 2.9 mW.

13.17 Crystal load capacitance (C10, C11)

ASSUMPTION: stray capacitance (PCB traces plus the STM32 OSC pins) of 2–5 pF.

```
Formula:       C_L = (C10 × C11) / (C10 + C11) + C_stray
Substitution:  C_L = (10 pF × 10 pF) / (20 pF) + 2…5 pF
Result:        C_L = 7–10 pF

Reverse form (to choose capacitors for a given crystal):
  C10 = C11 = 2 × (C_L − C_stray)   e.g. C_L = 8 pF, C_stray = 3 pF → 10 pF
```

Meaning: with 10 pF capacitors the crystal must be specified for a load capacitance of about 8 pF. A crystal specified for 12 pF or 18 pF would oscillate slightly fast, and USB timing depends on this frequency.

Open item: the crystal MPN, its specified C_L, ESR and shunt capacitance C0 are needed to check the oscillator gain margin against the STM32 HSE transconductance, as ST describes in application note AN2867.

13.18 VDDA ferrite filter (FB1, C6, C7)

ASSUMPTION: FB1 DC resistance 0.3 Ω; C7 equivalent series inductance 0.5 nH; FB1 low-frequency inductance 0.5–2 µH. All three require the actual MPNs.

```
DC drop:          ΔV = I_VDDA × R_DC = 5 mA × 0.3 Ω = 1.5 mV        (limit: VDDA = VDD)

Attenuation at 100 MHz (divider of Z_FB and Z_C7):
  |Z_C7| ≈ 2π × 100 MHz × 0.5 nH = 0.31 Ω
  A ≈ 20 × log10(0.31 Ω / 120 Ω) ≈ −52 dB

LC resonance:  f_r = 1 / (2π × √(L_FB × C_total)),  C_total = C6 + C7 = 1.1 µF
  L_FB = 0.5 µH → 215 kHz;   1 µH → 152 kHz;   2 µH → 107 kHz
```

Meaning: at the frequencies where digital noise lives, the bead and C7 attenuate noise by roughly 50 dB, while the DC drop is negligible. The filter has an LC resonance in the 100–200 kHz range, below the 340 kHz buck frequency. A ferrite's resistive behaviour normally damps that resonance, but the chosen MPN's impedance curve should be checked to confirm there is no peaking near 340 kHz.

13.19 SPI timing budget through the level translators

The ICM-20948 changes SDO on the falling edge of SCLK and the STM32 samples MISO on the next rising edge, half a clock period later. In that half period, the clock must cross U10 (3.3 V → 1.8 V), the IMU must drive SDO, and the data must cross U7 (1.8 V → 3.3 V).

```
t_round = t_pd,U10 + t_VD,SDO + t_pd,U7 = 4.84 ns + 59 ns + 5.07 ns = 68.9 ns
          (translator delays at 15 pF; SDO valid time at 20 pF)

Margin = T_SCLK / 2 − t_round − t_su,MISO(STM32)
```

SPI clock	Half period	Margin before MCU setup time	Assessment
7.00 MHz (IMU maximum)	71.4 ns	2.5 ns	Marginal
6.25 MHz (APB2 100 MHz / 16)	80.0 ns	11.1 ns	Acceptable
3.125 MHz (APB2 / 32)	160 ns	91 ns	Robust

13.20 Pull-up and pull-down currents

```
R16 (SPI1_CS pull-up), CS asserted:     I = 3.3 V / 10 kΩ = 0.33 mA
R1 as BOOT0 pull-down (recommended fix), button pressed: I = 3.3 V / 10 kΩ = 0.33 mA
```

Meaning: 10 kΩ is a standard compromise. It is low enough to hold a floating MCU pin at a defined level against leakage and noise, and high enough that the MCU drives against it with negligible current.

13.21 ESC signal series resistors

```
RC time constant:   τ = 22 Ω × 100 pF (ASSUMED wire + ESC input) = 2.2 ns
DShot600 bit time:  1 / 600 kbit/s = 1.67 µs → τ is 0.13 % of a bit
Short to GND:       I = 3.3 V / 22 Ω = 150 mA (> 25 mA STM32 per-pin maximum)
```

14. Conclusion

The current Version 1 PCB combines an STM32F411 controller, an ICM-20948 inertial sensor, multi-rail power conversion, logic-domain interfacing, USB, SWD, serial interfaces and four PWM-labelled outputs on a four-layer PCB.

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